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September 29, 1970
Interim Revision of
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00-G-682
8 August 1956

INTERIM FEDERAL SPECIFICATION

# GRINDING AND POLISHING MACHINE, DENTAL LABORATORY
## Bench Mounted, 115 Volt, 60 Cycle, AC

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification 00-G-682.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a machine suitable for the grinding and polishing operations necessary in Dental Prosthetic Laboratory.

1.2 Classification.

1.2.1 Grinding and Polishing Machine, Complete. The grinding and polishing machine shall be furnished complete with all accessories when specified (see 6.1).

1.2.2 Replacement Parts. Replacement parts for the grinding and polishing machine shall be of the following Classes, Types, and Styles, as specified (see 6.1).

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| | | |
|---|---|---|
| Class I. | | Chuck Adapter, Automatic, Right, |
| Class II. | | Chuck. |
| | Type A. | Automatic. |
| | Style 1. | Arbor Band, Right. |
| | Style 2. | Bur, Right. |
| | Style 3. | Tapered Spindle, Right. |
| | Style 4. | Stone, Right. |
| | Type B. | Non-Automatic. |
| | Style 1. | Tapered Spindle, Left. |
| | Style 2. | Stone, Left. |
| Class III. | | Spring, Replacement. |
| | Style 1. | Chuck Adapter. |
| | Style 2. | Bur Chuck. |

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards and drawing. The following specifications, standards, and drawings, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| CC-M-636 | Motors, Fractional Horse-Power (Alternating Current). |
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Boxes, Fiberboard. |
| PPP-B-676 | Box, Setup. |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-66 | Steel: Chemical Composition and Hardenability. |
| FED-STD-123 | Marking For Domestic Shipment (Civilian Agencies). |
| FED-STD-151 | Metals; Test Methods. |



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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-B-121 | Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-F-13924 | Finish, Black Oxide, for Iron and Steel. |
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible. |
| MIL-P-116 | Methods of Preservation. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

DRAWINGS

Defense Personnel Support Center (DPSC)

| | |
|---|---|
| 2039 | Adapter, Chuck, Dental Grinding and Polishing Machine, Right, Automatic. |
| 2040 | Chuck, Grinding and Polishing Machine, Dental Laboratory, Arbor Band, Right. |
| 2041 | Chuck, Grinding and Polishing Machine, Dental Laboratory, Bur, Right. |
| 2042 | Chuck, Grinding and Polishing Machine, Dental Laboratory, Tapered Spindle, Right. |
| 2043 | Chuck, Grinding and Polishing Machine, Dental Laboratory, Stone, Right. |



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| | |
|---|---|
| 22,001 | Chuck, Grinding and Polishing Machine, Dental Laboratory, Stone, Left. |
| 22,002 | Chuck, Grinding and Polishing Machine, Dental Laboratory, Tapered Spindle, Left. |
| 24,240 | Grinding and Polishing Machine, Bench Mounted. |
| 24,241 | Wiring Diagram, Grinding and Polishing Machine, Bench Mounted. |

(Copies of military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)*

ANSI C73.11-1966 Outlet Receptacles, Attachment Plug Caps and Appliance Plugs.

*Formerly the American Standards Association, and the United States of America Standards Institute, Inc. All A.S.A. Standards and USASI Standards are now designated as ANSI Standards.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N. Y. 10018).

UNDERWRITERS' LABORATORIES, INC.

*Underwriters' Laboratories Standards*

(Application for copies should be addressed to the Underwriters' Laboratories Inc., Walt Whitman Road, Melville, L.I., N.Y. 11749; 207 E. Ohio Street, Chicago, Illinois 60611; and 1655 Scott Blvd., Santa Clara, Calif. 95050).

3. REQUIREMENTS.

3.1 *Fire and casualty hazards.*

3.1.1 Prior to delivery, the contractor shall submit proof to the contracting agency, that the equipment he will supply under this specification conforms to the applicable requirements of the Underwriters' Laboratories, Inc., Standards. The label or listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the equipment conforms to this requirement.



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3.1.2 In lieu of the label or listing, the contractor may submit independent proof, satisfactory to the contracting agency, prior to delivery, that equipment conforms to the Underwriters' Laboratories, Inc., Standards.

3.1.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure the acceptance of his commodity.

3.2 Materials. Materials shall be new and shall conform to the applicable specifications and requirements specified herein. Materials not specifically designated shall be of high commercial quality throughout, and shall be entirely suitable for the intended purpose.

3.2.1 Steel. Steel used in the fabrication of the replacement part chucks and adapter and of the hardened motor shaft shall conform, respectively, to the chemical composition of AISI B1112 and AISI 1144 of FED-STD-66.

3.3 Style and Design. Style and design of the complete grinding and polishing machine and its replacement parts shall be as shown in Defense Personnel Support Center (DPSC) Drawing 24240 and in Drawings referenced in 3.4.1.2.1 to 3.4.1.2.9. Machine shall be suitable for operation on 120 volt, 60 cycle, single phase alternating current and shall be connected together, electrically, as shown in DPSC Drawing 24241. Machine shall be designed to insure rigid, rugged, and durable construction and shall be suitable for grinding and polishing operations in dental prosthetic procedures. The machine shall weigh 37 $\pm$ 3 pounds.

3.3.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the insert trays.

3.4 Construction. The complete grinding and polishing machine shall consist of the motor, chuck removing device, connecting cord, motor base, control switch and the adapter, chuck and spring component parts which shall be furnished in the quantities shown in table I.



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Table I. Basic Machine and Component Parts

| Item | Component Parts | Quantity | Reference |
|---|---|---|---|
| Basic Machine Complete | | 1 | 3.4.1 through 3.4.1.2.9 |
| | Chuck Adapter | 1 | 3.4.1.2.1 |
| | Chuck, Automatic | | |
| | Arbor Band, Right | 1 | 3.4.1.2.2 |
| | Bur, Right | 1 | 3.4.1.2.3 |
| | Tapered Spindle, Right | 1 | 3.4.1.2.4 |
| | Stone, Right | 1 | 3.4.1.2.5 |
| | Chuck, Non-Automatic | | |
| | Tapered Spindle, Left | 1 | 3.4.1.2.7 |
| | Stone, Left | 1 | 3.4.1.2.6 |
| | Spring, Replacement | | |
| | Chuck Adapter | 4 | 3.4.1.2.1 and 3.4.1.2.8 |
| | Bur, Chuck | 4 | 3.4.1.2.3 and 3.4.1.2.9 |

3.4.1 Grinding and Polishing Machine, Complete.

3.4.1.1 Motor. The motor shall consist of the integral components specified below and shall conform to the applicable requirements of CC-M-636, for type I, class 1b, style B1, continuous duty or intermittent duty at load, no load and rest. Motor furnished shall be a split-phase, ball bearing type, 1/6 HP, 115 volt, 1725/3450 revolutions per minute (R.P.M.), 60 cycle, single phase, alternating current unit. The motor shall operate with a minimum of noise at all loads and speeds and shall withstand, without failure, the dielectric strength test specified in 4.3.2.

3.4.1.1.1 Frame. The motor frame shall be a ring-shaped steel band which shall serve as the central medium about which the motor components are assembled. The frame shall have sufficient sectional area and depth around its outer periphery to provide for the reception of the laminated stator core and to withstand any distortion caused by the press fit insertion of the stator core. The tightness of this press fit shall be further implemented by installing the main (high and low speed) windings and starting windings on the newly completed stator core, pressing core in frame, and then subjecting the assembly to the varnish dipping and baking treatment given electrical units. The frame shall also provide in combination with the stator core, the passages which are on a 45 degree radial line in each quadrant and which accommodate the through bolts used to fasten the end bells to the frame. The frame shall be recessed at both outer edges around the band circle to produce its portion of the rabbeted joint formed with the end bells. The frame shall be drilled to accommodate the bolts used to fasten this assembly to the separate motor base where this type of base design is used.



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3.4.1.1.2 Stator. The motor stator shall be fabricated by riveting or welding together a stack of 25 gauge (.022 inch) electric sheet steel laminations to form a laminated core structure. The laminations shall be in the form of single circular punchings which, after being punched out, shall be heat treated to produce the oxide coating necessary to maintain a laminated rather than a solid core structure. The inner periphery of the laminations shall be furnished with slots in which the main windings and auxiliary starting windings shall be installed. The starting windings shall be displaced 120 electrical degrees from the main windings, and shall be connected in parallel with the main windings across the power supply lines during all starting operations. The electrical interconnection of the main and starting windings with the double pole, double throw (or single pole, triple throw) switch, as shown in DPSC Drawing 24241, shall provide the means of converting the motor from high to low grinding speed when this action is desired. An alternate motor (stator core) design (see DPSC Drawing 24240) shall involve riveting or welding the laminations into an octagonally shaped, laminated core structure which shall be press-fitted into a frame or field ring. The passages for the through bolts, which fasten the end bells to the ring assembly, shall be provided between the field ring and sloping octagonal facings on the 45 degree radial lines in each quadrant. The press fit of the stator core in the field ring shall be implemented by cast studs (see 3.4.1.1.5), on the end bells, which shall grip the stator core and hold it in proper alignment.

3.4.1.1.3 Rotor. The rotor shall be constructed by building up a cylindrically shaped core of 25 gauge sheet steel laminations on a dummy shaft and pouring an aluminum casting around them in a mould designed to produce a squirrel cage assembly. The outer periphery of the rotor shall be machined, the dummy shaft removed, and the inner periphery machined to accommodate a press fit of the permanent motor shaft. The rotor assembly shall be heat treated and quenched after machining to improve its magnetic qualities and to insure the formation of the laminated core structure. A centrifugal switch shall be installed on one end of the rotor housing to open the auxiliary starting windings as the motor reaches a speed of 1300-1400 R.P.M. and the need for the auxiliary fields starting capability is ended. The rotor shall be accurately balanced, statically and dynamically, to provide vibration-less operation at each speed.

3.4.1.1.4 Shaft. The motor shaft shall extend beyond each end of the motor, shall be accurately ground to precision limits, and shall be provided with tapered ends for the accommodation of standard lathe chucks. The exposed length of shaft, beyond end of chuck remover, shall be 1 $\pm$ 1/16 inch. The shaft taper shall be 0.250 $\pm$ 0.001 inch per foot, and the shaft shall end with a diameter of 0.355 $\pm$ 0.001 inch at the extreme outer ends. The shaft shall not move parallel to its long axis, except when chucks are being removed. The center of the shaft shall be no less than 4 inches and no more than 5 inches from the surface of work bench. The sections of shaft adjacent to the rotor shall be suitably machined to the diameters and shaft lengths required for the reception of the bearing and chuck remover assemblies and for the auxiliary starting winding centrifugal switch.



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3.4.1.1.5 End Bells. The motor end bells shall serve the dual purpose of enclosing the open ends of the motor assembly and of providing a housing for the ball bearings and other shaft accessories. The end bells shall be notched along their outer periphery to provide their portion of the rabbeted joint formed with the motor frame. The end bells shall be drilled adjacent to their outer periphery, and on a 45 degree radial line in each quadrant, for the reception of the through bolts used to fasten the motor assembly together. The end bells shall be provided with suitably machined openings at their centers for the passage of the rotor shaft and with the housing wells required for the ball bearing and chuck remover assemblies. The end bells shall also be drilled and tapped, adjacent to their inner periphery and on the 45 degree radial lines in each quadrant, for the bolts used to fasten the chuck removers to the motor assembly. On the alternate motor design referenced in 3.4.1.1.2, the male part of the chuck remover assembly shall be held in position inside the end bell hub by a set screw. The male part shall be threaded externally, on the opposite end, for the attachment of the female (lever handle) part of the chuck remover assembly. Also on the alternate motor design, the end bells shall be furnished with two cast studs (feet) in each quadrant, which shall be located along the inner edge of their outer peripheries.

3.4.1.1.6 Bearings. Motors shall be furnished with ball bearings of suitable design, size, and quality. The bearings shall have permanent lubrication sealed in so that no further lubrication is required for the life of the bearings. Ball bearings shall be accurately fitted into the motor and shall be readily and easily removable. Bearings shall be standardized to permit replacement without machining or fitting. Washers shall be provided at the outer end of each bearing.

3.4.1.1.7 Chuck removing device. A suitable chuck removing device shall be located at each end of the shaft to provide easy removal of chuck by turning of lever handle. The chuck removing device shall be designed so that its operation does not exert excess thrust (or chuck load) between the balls and races of the bearing. The device shall provide adequate clearance for the free running motor shaft and shall be equipped with a gasket to keep out dust or other foreign matter.

3.4.1.1.8 Connecting cord. The power connecting cord for motor shall consist of a rubber covered service cable with a steel armored connector plug attached. The power cord shall be a 3-conductor cable equipped with a 3-prong plug suitable for use with receptacles specified in American National Standards Institute (ANSI) Standard ANSI C73.11-1966, for Caps and Receptacles, 125 Volts, 15 Ampere, 2-Pole, 3-Wire Grounding Type. The current rating of the cord shall be no less than that of the motor. External connecting cord shall be no less than 8 feet in length.



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3.4.1.1.9 Motor Base. The motor shall be supported by a base, cast integrally, or separately fastened to the motor frame. The base shall have sufficient metallic thickness and area to provide firm support for the motor and shall have adequate weight to forestall slippage during operation. The base shall have four feet, each equipped with heavy rubber pads to further assist in control of slippage and to prevent damage to bench top. The base shall serve as a housing for the motor control switch.

3.4.1.1.10 Control switch. The control switch shall be totally enclosed, shall be located in the front of the motor base, and shall be a three position, snap-action toggle or lever type. Adequate designation of switch position shall be provided for "off", "high speed", and "low speed" positions.

3.4.1.2 Replacement Parts. Replacement parts, when specified, shall be furnished as described in 3.4.1.2.1 through 3.4.1.2.9 and in the quantities specified in table I.

3.4.1.2.1 The class I adapter shall be for use on the right hand side of the machine, and shall permit the interchangeability of adapter-type dental chucks while the machine is in operation. The adapter shall have three Allen screws for attachment to the machine shaft and for adjustment for true-running. Each adapter shall be furnished with four extra springs. The dimensions, style, and design of the adapter shall conform to DPSC Drawing 2039. Adapter shall be fabricated of material as specified in this drawing. Adapter shall have proper gripping and holding ability with no radial slippage, when tested as specified in 4.3.3.1.

3.4.1.2.2 The class II, type A, style 1 chuck shall be fabricated of carbon steel, shall be used for emery arbor band and shall be suitable for use with the automatic adapter as specified in 3.4.1.2.1. The dimensions, style, and design of the chuck shall conform to DPSC Drawing 2040.

3.4.1.2.3 The class II, type A, style 2 chuck shall be fabricated of carbon steel, shall be used for burs, mandrels, and mandrel mounted silicon carbide points and wheels, and shall be suitable for use with the automatic adapter as specified in 3.4.1.2.1. The dimensions, style, and design of the chuck shall conform to DPSC Drawing 2041. Chuck shall be furnished with four extra springs. The bur chuck shall have proper gripping and holding ability with no radial slippage, when tested as specified in 4.3.3.2.

3.4.1.2.4 The class II, type A, style 3 chuck shall be fabricated of carbon steel, shall be a tapered spindle used for cloth, felt and brush wheels, and shall be suitable for use with the automatic adapter as specified in 3.4.1.2.1. The dimensions, style, and design of the chuck shall conform to DPSC Drawing 2042.



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3.4.1.2.5 The class II, type A, style 4 chuck shall be fabricated of carbon steel, shall be used for unmounted stone or rubber lathe wheels, and shall be suitable for use with the automatic adapter as specified in 3.4.1.2.1. The dimensions, style, and design shall conform to DPSC Drawing 2043.

3.4.1.2.6 The class II, type B, style 2 chuck shall be fabricated of brass or bronze, shall be used for unmounted stone or rubber lathe wheels, and shall be suitable for attachment to the left hand side of the machine. The dimensions, style, and design shall conform to DPSC Drawing 22001.

3.4.1.2.7 The class II, type B, style 1 chuck shall be fabricated of brass or bronze, shall be a tapered spindle used for cloth, felt and brush buffing wheels and shall be suitable for attachment to the left hand side of machine. The dimensions, style, and design shall conform to DPSC Drawing 22002.

3.4.1.2.8 The class III, style 1 replacement spring shall be suitable for use with the class I chuck adapter, and shall conform in design, style and dimensions to Defense Personnel Support Center (DPSC) Drawing 2039.

3.4.1.2.9 The class III, style 2 replacement spring shall be suitable for use with the class II, type A, style 2 chuck and shall conform in design, style, and dimensions to DPSC Drawing 2041.

3.5 *Dimensions.* The *dimensions* of the machine, in inches, shall be as follows:

| | |
|---|---|
| Overall height | 7-3/4 ± 1/2 (without feet) |
| Overall length of shaft | 14-1/2 ± 1 |
| Diameter of housing | 6-1/2 ± 1/2 |
| Width of base | 7 ± 1/2 |
| Depth of base | 8 ± 1/2 |

3.6 Finish. Machine shall have a black enamel or hammertone gray finish, except for the chuck remover which shall be nickel or chromium plated. The shaft shall be neither plated nor painted. Chucks and adapter fabricated of carbon steel shall have a black oxide finish conforming to class 1 of MIL-F-13924. Chucks fabricated of brass or bronze shall have a lacquered, satin finish. The brass screw of stone chuck, right, and arbor band, right, shall have a black oxide finish. The accessory replacement springs shall have a uniform, deep colored, blue temper.

3.7 Identification marking. Each machine shall be provided with a readily visible metal nameplate permanently affixed thereto. The plate shall be marked with the name or registered trademark of the manufacturer, model designation, serial number, current, voltage and frequency. The markings on the plate shall be clear and permanent, and shall be accomplished by die stamping into the surface of the metal. Alternate methods of marking the nameplate (other than decalcomania) may be submitted to the contracting officer



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for approval. As an alternate method, the required marking shall be provided on two metal plates permanently affixed, one to the motor housing and the other to the base of the lathe.

3.7.1 Identification marking for replacement parts. The replacement parts shall be permanently and legibly marked with the name or registered trademark of the manufacturer. Marking shall appear on a major outside surface of each accessory in an area where it would not adversely affect proper functioning.

3.8 Tolerances. Unless otherwise specified herein, tolerances on dimensions shall be ± 5 percent, providing this does not sacrifice function and first quality workmanship.

3.9 Performance. The machine shall be suitable for grinding and polishing operations necessary in dental prosthetic laboratory procedures, when tested as specified in 4.3.3.

3.10 Service data. The contractor shall furnish two copies of a service data manual with each unit. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. This service data shall contain complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only.

3.10.1 Approval of service data. As soon as practicable after award of the contract or purchase order, the contractor shall furnish two preliminary copies of his proposed service data to the contracting officer, via the cognizant Government inspector, for review and comment by Directorate of Medical Material, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Material, Defense Personnel Support Center (based on a maximum period of 30 days for such review), and to allow the contractor to incorporate required revisions and corrections in the final service data without delaying delivery under the contract or purchase order.

3.10.2 Waiver of review and distribution copies of manual. Should the supplier have furnished manuals which have been reviewed, commented on, and approved by the Directorate of Medical Material, for the identical item, within two years preceding the date of contract, and should the supplier propose to furnish manuals identical to those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.



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3.11 Workmanship. Workmanship shall be first class throughout. The grinding and polishing machine shall be free from defects which detract from its appearance or may impair its serviceability. Motor components shall be free from sharp or rough edges, burs, nicks, and slivers to preclude damage to wire insulation. External surfaces shall be dust-proof, drip-proof, and shall be free from scale, grinding marks, and ridges to permit easy cleaning of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Governments' request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the grinding and polishing machine, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the lot number, lot size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations required to assure conformance of those machines or lots of machines to be offered are classified as specified in tables II (see 4.2.1), and III (see 4.2.2). The tests required for acceptance are specified in table IV (see 4.2.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. The unit of product for examination purposes shall be one grinding and polishing machine or individual component when ordered as a replacement item.



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Table II. Inspection and Acceptable Quality Levels

| Type defect | Inspection level | AQL (percent defective) |
|---|---|---|
| Major A | II | 1.0 |
| Major B | S-2 | 1.5 |
| Minor A | II | 2.5 |
| Minor B | S-2 | 4.0 |

4.2.2 Examination. The machines shall be examined for the defects listed in table III. Examination shall include, but shall not be limited to, those defects shown in the table.

Table III. Classification of Defects

| Category | Defect |
|---|---|
| Critical | None defined |
| Major A | |
| 101 | Quantity of adapter and chuck springs not as specified. |
| 102 | Sharp edges, burrs, and slivers present on stator assembly. |
| 103 | Motor components not properly assembled. |
| 104 | Welding of stator assembly of poor quality. |
| 105 | Plating of chuck remover shows signs of peeling or flaking. |
| 106 | Design not as specified. |
| 107 | Construction not as specified. |
| 108 | Centrifugal switch not cutting out properly. |
| Major B | |
| 151 | Stator assembly has poorly machined, surface finish. |
| 152 | Dimensions affecting fit of chucks not as specified. |
| 153 | Taper of motor shaft not as specified. |
| 154 | Bur chuck springs show signs of corrosion. |
| 155 | Threads missing on throat of arbor band expansion screw. |
| Minor A | |
| 201 | Finish of chuck and adapter components not as specified. |
| 202 | Identification marking incorrect, incomplete, or illegible. |
| 203 | Packaging not as specified. |
| 204 | Packing not as specified. |
| 205 | Nameplate not permanently attached to machine. |
| Minor B | |
| 251 | Dimensions (other than 152 and 153) not as specified. |
| 252 | Motor cable insulation not as specified. |
| 253 | Color of machine finish not as specified. |
| 254 | Set of instructions missing from machine. |



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4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table IV, except that the acceptance number shall be zero for all sample sizes. The unit of product for testing purposes shall be one grinding and polishing machine or individual component when ordered as a replacement item.

Table IV. Sampling and Quality Conformance Criteria for Testing

| Characteristic | Requirement | Method | Inspection level |
|---|---|---|---|
| Composition | 3.2.1 | 4.3.1 | * |
| Dielectric strength | 3.4.1.1 | 4.3.2 | 100% |
| Performance | 3.9 | 4.3.3 | S2 |

* Sufficient amounts from each lot of each metal used shall be selected for chemical analysis.

4.3 Test methods. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Chemical analysis. Metal analysis of the chucks and adapter shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of conflict, method 111 shall govern.

4.3.2 Dielectric strength test. All electrical components shall be tested after complete assembly and shall withstand without failure a dielectric strength test of twice the rated voltage plus 1000 volts, or a total of 1250 volts applied for one minute. This test shall be made on each of the individual lines of the circuit, using the metal frame as the other leg or ground of the 1250 volt test.

4.3.3 Performance test. The grinding and polishing machine shall be given a preliminary running test by connecting the unit to the proper electric current source and operating the control switch in successive tests, in the "low" and "high" speed positions. A direct reading tachometer shall be used to check for each test position that: (1) the motor accelerates properly; (2) the centrifugal switch cuts out at specified speed value; and (3) final running speed is obtained conforming to control switch indication. The chucks shall be given a practical performance test by evaluating the ability of the chuck to properly hold the various rotating instruments for which they are designed, in a prescribed dental prosthetic procedure.



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4.3.3.1 Grip test, adapter. Each assembled adapter shall be mounted on a suitable power-driven shaft and rotated at 200 to 300 rpm. Hardened steel test pieces of 0.3733 to 0.3735 inch diameter and 0.359 to 0.360 inch diameter shall be inserted into the adapter and tested for the requirements specified in 3.4.1.2.1. The test pieces shall have a knob or bushing 5/8 to 3/4 inch in diameter securely fastened to the free end for hand grip.

4.3.3.2 Grip test, bur chuck. Each assembled bur chuck shall be inserted in an adapter mounted on a suitable power driven shaft and rotated at 200 to 300 rpm. Hardened steel test pieces of 0.0939 to 0.0941 inch diameter and 0.084 to 0.085 inch diameter shall be inserted into the bur chuck and tested for the requirements specified in 3.4.1.2.3. The test pieces shall have a knob or bushing 5/8 to 3/4 inch in diameter securely fastened to the free end for hand grip.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One machine with replacement parts, or one chuck, or one spring, of the stock number specified, constitutes one unit of issue. Four extra springs shall be furnished with each unit of Stock Numbers 6520-539-0155 and 6520-538-9850.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

| Federal stock number | Class | Type | Style | Unit package | Intermediate package | Exterior container |
|---|---|---|---|---|---|---|
| | see 1.2.1 | see 1.2.1 | see 1.2.1 | Not req'd | Not req'd | 1 unit |
| 6520-526-0131 | II | B | 1 | 1 unit | 6 units | 288 units |
| 6520-538-9850 | I | - | - | 1 unit | 6 units | 72 units |
| 6520-539-0075 | II | A | 1 | 1 unit | 6 units | 144 units |
| 6520-539-0155 | II | A | 2 | 1 unit | 6 units | 144 units |
| 6520-539-0305 | II | A | 3 | 1 unit | 6 units | 144 units |
| 6520-539-0350 | II | B | 2 | 1 unit | 6 units | 96 units |
| 6520-539-0550 | II | A | 4 | 1 unit | 6 units | 144 units |
| 6520-539-1200 | III | - | 1 | 1 unit | 12 units | 288 units |
| 6520-539-1225 | III | - | 2 | 1 unit | 12 units | 288 units |



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5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit preparation (Machine with Replacement Parts). Each unit shall be thoroughly cleaned and dried in accordance with the applicable method, as outlined in MIL-P-116. Exterior base metal surfaces subject to corrosion shall be treated with a P-3 preservative, as specified in MIL-P-116. All exterior finished surfaces shall be protected by wrapping with neutral paper. Work shall be accomplished in such a manner as to insure against marring, scratching, or other damage to the unit. Replacement parts shall be packaged as specified in 5.1.3.2.

5.1.3.2 Unit package (Replacement Parts). Each unit shall be thoroughly cleaned and dried in accordance with the applicable method, as outlined in MIL-P-116. Each unit shall be preserved with type P-9 preservative and shall be wrapped in greaseproof barrier material conforming to MIL-B-121. Each unit shall be further packaged in accordance with MIL-P-116, method IA-8.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size and design, constructed in accordance with PPP-B-566, or PPP-B-676, except that commercial colors will be accpetable. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container (Machine with Replacement Parts). Exterior container shall be designed for a type 3 load and shall be constructed in accordance with PPP-B-601, overseas type; or PPP-B-621, class 2. The unit shall be adequately secured within the exterior container, by means of bolts, wood braces, blocks, or other suitable devices, to insure against shifting and breakage, or marring of surfaces. Closure and strapping shall be in accordance with the appendix of applicable box specification.



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5.2.1.2 Exterior container (Replacement Parts). Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.3 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container (Machine with Replacement Parts). Each unit shall be packed as specified in 5.2.1.2 except that the boxes shall be of the domestic type, and the case liner is not required.

5.2.2.2 Exterior container (Replacement Parts). Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery, by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.



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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Complete grinding and polishing machine or individual replacement parts, as required (see 1.2.1 and 1.2.2).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 This specification does not include all types, classes, grades or sizes, of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1074

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

00-G-00682a(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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**CAUTION**

**NOTICE TO BIDDERS/OFFERORS**

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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